

Yingfan (Ivan) Luo

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EDUCATION

University of British Columbia

Vancouver, BC

Bachelor of Applied Science in Computer Engineering — GPA: 3.9/4.0 (87%)

Expected December 2028

Relevant Coursework: Computing Systems, Software Construction, Data Structures and Algorithms, Computer Architecture, Operating Systems

EXPERIENCE

Apera AI Inc.

Sept. 2026 – Dec. 2026

Software Engineer Intern

Vancouver, BC

- Incoming co-op on the **C++** software team building 4D vision software for industrial robotics.

Ruboss Technology Corporation (Leanpub)

May 2026 – Aug. 2026

Software Engineer Intern

Vancouver, BC

- Shipped Leanpub's first native iOS app to the App Store in **7 weeks** as one of two engineers: **105 screens** covering reading, book publishing, pricing, and sales, built in **SwiftUI** over a **GraphQL** API; **1,000+ downloads**.
- Built real-time collaborative editing so co-authors can write a book together with live sync across web and iOS, using **CRDTs** (Yjs) over Rails **ActionCable**; ported Yjs's binary sync protocol to **Swift**.
- Built the app's two chapter-writing editors, a source editor for power users and a simplified prose editor for first-time authors, enabling authors to write entire books from an iPhone (**UIKit 2**).
- Built an **AI catalog assistant** for leanpub.com that answers natural-language requests like "find me a book on X", powered by a tool-calling **LLM** pipeline over the live catalog.
- Launched **GitHub-backed books**: authors connect a repository and Leanpub publishes from it, with automated repo provisioning, ownership transfer, and hardened, injection-safe input validation.

UBC Rocket - Thrust Vector Control

Sept. 2025 – Present

Embedded Software Engineer

Vancouver, BC

- Built the flight team's ground control station in **C++** and **Qt**, rendering 100+ Hz telemetry with live 3D attitude visualization and satellite-view map.
- Designed non-blocking **STM32** firmware (SPI) delivering a jitter-free **1 kHz** IMU data pipeline, meeting the timing budget the attitude-control loop depended on.
- Built a bidirectional radio link with **COBS**-encoded message framing and an updated **Protobuf** schema, plus CSV packet logging that enabled post-flight analysis of command and telemetry data.
- Implemented the controls team's **PID** flight-control law from spec and built a ground-station tuning page with saved gain presets; guarded by a 30-test **CTest** regression harness.
- Developed and tuned **ESC/motor-control firmware** and built a **Python** data-collection pipeline measuring force/torque vs. thrust %, enabling thrust-performance validation and tuning.

PROJECTS

Live Market Data FPGA Processor | *SystemVerilog, Python, UART*

- Built a real-time data pipeline on a **DE10-Lite FPGA**, streaming live binary packets from a **Python** WebSocket feed over **UART** into a **SystemVerilog** parser FSM.
- Implemented hardware signal generation and cycle-level **latency measurement (50 MHz)**, reporting timing results to a live PC dashboard for analysis.

Unix Shell | *C, Linux*

- Built a Unix shell in **C** with a REPL supporting foreground/background job execution via **fork/execve** and process groups.
- Implemented **job control** (**fg**, **bg**, **kill**, **jobs**) and POSIX signal handlers (**SIGINT**, **SIGTSTP**, **SIGCHLD**), eliminating zombie processes.

TECHNICAL SKILLS

Programming Languages: C, C++, Java, JavaScript, TypeScript, Python, Ruby, Swift, SystemVerilog, ARM Assembly

Frameworks: React, Next.js, Remix, Node.js, SwiftUI, Ruby on Rails, GraphQL, Qt, Protobuf

Developer Tools: Git/GitHub, Claude Code, Xcode, Linux, CTest